Assignment No. 8: Coverage Generation and Analysis using Synopsys VCS

Q1. Perform functional coverage analysis for a 4-bit Universal Shift Register using Synopsys VCS.
a) Develop a testbench that verifies all operating modes.
b) Enable Line, Toggle, Branch, and Condition coverage during compilation.
c) Generate the coverage database and coverage report.
d) Analyze the generated report and identify any uncovered statements

- ## Testbench:

```
`timescale 1ns/1ps
module tb_universal_shift_register;
    reg clk, rst;
    reg [1:0] mode;
    reg [3:0] parallel_in;
    reg serial_in_left, serial_in_right;
    wire [3:0] q;
    universal_shift_register uut (
        .clk(clk),
        .rst(rst),
        .mode(mode),
        .parallel_in(parallel_in),
        .serial_in_left(serial_in_left),
        .serial_in_right(serial_in_right),
        .q(q)
    );
    always #5 clk = ~clk;
    initial begin
        $dumpfile("universal_shift_register.vcd");
        $dumpvars(0, tb_universal_shift_register);

        clk = 0;
        rst = 1;
        mode = 2'b00;
        parallel_in = 4'b0000;
        serial_in_left = 0;
        serial_in_right = 0;
        #12 rst = 0;
        // Parallel load: Q = 1010
        @(negedge clk);
        mode = 2'b11;
        parallel_in = 4'b1010;
```

```
        // Hold
        @(negedge clk);
        mode = 2'b00;

        // Shift right
        @(negedge clk);
        mode = 2'b01;
        serial_in_left = 1;

        // Shift left
        @(negedge clk);
        mode = 2'b10;
        serial_in_right = 1;

        // Load a different value
        @(negedge clk);
        mode = 2'b11;
        parallel_in = 4'b1100;

        // Hold again
        @(negedge clk);
        mode = 2'b00;

        repeat (2) @(negedge clk);
        $finish;
    end

    initial begin
        $monitor("Time=%0t rst=%b mode=%b parallel_in=%b q=%b",
                 $time, rst, mode, parallel_in, q);
    end
endmodule
```

- **Commands:**

vcs -cm line+tgl+branch+cond -o Output/univ_shift_reg.out ../RTL/Design/universal_shift_register.v ../RTL/TB/tb_universal_shift_register.v

./univ_shift_reg.out -cm line+tgl+branch+cond

urg -dir univ_shift_reg.out.vdb/ -report ../Coverage/

- **Coverage Report:**

```
Time=25000 rst=0 mode=11 parallel_in=1010 q=1010
Time=30000 rst=0 mode=00 parallel_in=1010 q=1010
Time=40000 rst=0 mode=01 parallel_in=1010 q=1010
Time=45000 rst=0 mode=01 parallel_in=1010 q=1101
Time=50000 rst=0 mode=10 parallel_in=1010 q=1101
Time=55000 rst=0 mode=10 parallel_in=1010 q=1011
Time=60000 rst=0 mode=11 parallel_in=1100 q=1011
Time=65000 rst=0 mode=11 parallel_in=1100 q=1100
Time=70000 rst=0 mode=00 parallel_in=1100 q=1100
$finish called from file "../RTL/TB/tb_univ_shift_reg.v", line 66.
$finish at simulation time                90000

------------------------------------------------------------------------
VCS Coverage Metrics: during simulation line, cond, branch, tgl was monitored
------------------------------------------------------------------------

 Coverage status: End of All Coverages ...

          V C S   S i m u l a t i o n   R e p o r t
Time: 90000 ps
CPU Time:      0.270 seconds;       Data structure size:   0.0Mb
Thu Oct 22 20:25:02 2026
[SCL] 10/22/2026 20:25:02 PID:302143 Client:smartlab6.calicut.nielit.in checkin (null)
[oitp_b2-13@smartlab6 Output]$
```

SYNOPSYS

Dashboard

dashboard | hierarchy | modlist | groups | tests | asserts

Date: Thu Oct 22 20:26:14 2026
User: oitp_b2-13
Version: W-2024.09
Command line: urg -dir univ_shift_reg.out.vdb/ -report ../Coverage/
Number of tests: 1

Total Coverage Summary

SCORE	LINE	COND	TOGGLE	BRANCH
84.21	97.87		71.43	83.33

Hierarchical coverage data for top-level instances

SCORE	LINE	COND	TOGGLE	BRANCH	NAME
84.21	97.87		71.43	83.33	tb_universal_shift_register

Total Module Definition Coverage Summary

SCORE	LINE	COND	TOGGLE	BRANCH
84.21	97.87		71.43	83.33

Module : tb_universal_shift_register

SCORE	LINE	COND	TOGGLE	BRANCH
85.71	100.00		71.43	

Source File(s) :
/home/oitp_b2-13/Desktop/Shashaank_Malviya/Simulation/../RTL/TB/tb_univ_shift_reg.v

Module self-instances :

NAME	SCORE	LINE	COND	TOGGLE	BRANCH
tb_universal_shift_register	85.71	100.00		71.43	

Module Instance : tb_universal_shift_register

Instance :

SCORE	LINE	COND	TOGGLE	BRANCH
85.71	100.00		71.43	

Instance's subtree :

SCORE	LINE	COND	TOGGLE	BRANCH
84.21	97.87		71.43	83.33

Line 〉 Toggle 〉

Toggle Coverage for Module : tb_universal_shift_register

	Total	Covered	Percent
Totals	7	2	28.57
Total Bits	28	20	71.43
Total Bits 0->1	14	12	85.71
Total Bits 1->0	14	8	57.14
Signals	7	2	28.57
Signal Bits	28	20	71.43
Signal Bits 0->1	14	12	85.71
Signal Bits 1->0	14	8	57.14

Signal Details

Name	Toggle	Toggle 1->0	Toggle 0->1
clk	Yes	Yes	Yes
rst	No	Yes	No
mode[1:0]	Yes	Yes	Yes
parallel_in[0]	No	No	No
parallel_in[1]	Yes	Yes	Yes

Q2. Perform functional coverage analysis for an 8-bit Arithmetic Logic Unit (ALU) using Synopsys VCS.
a) Develop test cases to verify all arithmetic and logical operations.
b) Generate Line, Toggle, Branch, and Condition coverage reports.
c) Identify operations that are not fully covered.
d) Modify the testbench to improve the overall functional coverage.
e) Compare the initial and updated coverage reports.

- ## Initial Testbench:

```
`timescale 1ns/1ps
module tb_alu_8bit_init;

    reg [7:0] A, B;
    reg [2:0] opcode;
    wire [7:0] result;
    alu_8bit uut (
        .A(A),
        .B(B),
        .opcode(opcode),
        .result(result)
    );
    initial begin
        $dumpfile("../VCD/alu_8bit_init.vcd");
        $dumpvars(0, tb_alu_8bit_init);

        // Addition
        A = 8'd20; B = 8'd10; opcode = 3'b000;
        #10;
        // AND
        A = 8'hF0; B = 8'h0F; opcode = 3'b010;
        #10;
        // XOR
        A = 8'hAA; B = 8'hFF; opcode = 3'b100;
        #10;

        // Left shift
        A = 8'h0F; B = 8'd2; opcode = 3'b101;
        #10;

        $finish;
    end
    initial begin
        $monitor("Time=%0t A=%h B=%h opcode=%b result=%h",
                 $time, A, B, opcode, result);
    end
endmodule
```

- **Initial Coverage:**

```
7020@14.139.1.126 Checkout succeeded VCS-BASE-RUNTIME 2024.09 from VCS-Base-Run
ime-Pkg

 VCS Coverage Metrics Release W-2024.09 Copyright (c) 1991-2024 by Synopsys Inc
Time=0 A=14 B=0a opcode=000 result=1e
Time=10000 A=f0 B=0f opcode=010 result=00
Time=20000 A=aa B=ff opcode=100 result=55
Time=30000 A=0f B=02 opcode=101 result=3c
$finish called from file "../RTL/TB/tb_alu_8bit_init.v", line 36.
$finish at simulation time                40000

------------------------------------------------------------------------
VCS Coverage Metrics: during simulation line, cond, branch, tgl was monitored
------------------------------------------------------------------------

 Coverage status: End of All Coverages ...

          V C S   S i m u l a t i o n   R e p o r t
Time: 40000 ps
CPU Time:      0.270 seconds;       Data structure size:   0.0Mb
Thu Oct 22 20:47:56 2026
[SCL] 10/22/2026 20:47:56 PID:312758 Client:smartlab6.calicut.nielit.in checkin
(null)
[oitp_b2-13@smartlab6 Output]$
```

SYNOPSYS®

Dashboard

dashboard | **hierarchy** | **modlist** | groups | **tests** | asserts

Date: Thu Oct 22 20:48:41 2026
User: oitp_b2-13
Version: W-2024.09
Command line: urg -dir alu_8bit_init.out.vdb/ -report ../Coverage
Number of tests: 1

Total Coverage Summary

SCORE	LINE	COND	TOGGLE	BRANCH
68.55	85.29		75.93	44.44

Hierarchical coverage data for top-level instances

SCORE	LINE	COND	TOGGLE	BRANCH	NAME
68.55	85.29		75.93	44.44	tb_alu_8bit_init

Total Module Definition Coverage Summary

SCORE	LINE	COND	TOGGLE	BRANCH
68.55	85.29		75.93	44.44

- **Unchecked operations:**

Module : alu_8bit

SCORE	LINE	COND	TOGGLE	BRANCH
56.79	50.00		75.93	44.44

Source File(s) :
/home/oitp_b2-13/Desktop/Shashaank_Malviya/Simulation/../RTL/Design/alu_8bit.v

Module self-instances :

NAME	SCORE	LINE	COND	TOGGLE	BRANCH
tb_alu_8bit_init.uut	56.79	50.00		75.93	44.44

Module Instance : tb_alu_8bit_init.uut

Instance :

SCORE	LINE	COND	TOGGLE	BRANCH
56.79	50.00		75.93	44.44

Instance's subtree :

SCORE	LINE	COND	TOGGLE	BRANCH
56.79	50.00		75.93	44.44

Line 〉 Toggle 〉 Branch 〉

Line Coverage for Module : alu_8bit

	Line No.	Total	Covered	Percent
TOTAL		10	5	50.00
ALWAYS	10	10	5	50.00

```
9                           always @(*) begin
10          1/1                 case (opcode)
11          1/1                     3'b000: result = A + B;       // Addition
12          0/1     ==>             3'b001: result = A - B;       // Subtraction
13          1/1                     3'b010: result = A & B;       // Bitwise AND
14          0/1     ==>             3'b011: result = A | B;       // Bitwise OR
15          1/1                     3'b100: result = A ^ B;       // Bitwise XOR
16          1/1                     3'b101: result = A << B[2:0]; // Logical left shi
17          0/1     ==>             3'b110: result = A >> B[2:0]; // Logical right shi
18          0/1     ==>             3'b111: result = 8'b00000000; // Unused opcode
19          0/1     ==>             default: result = 8'b00000000;
```

Toggle Coverage for Module : alu_8bit

	Total	Covered	Percent
Totals	4	0	0.00

- **Updated Testbench:**

```
`timescale 1ns/1ps
module tb_alu_8bit_updated;

    reg [7:0] A, B;
    reg [2:0] opcode;
    wire [7:0] result;

    alu_8bit uut (
        .A(A),
        .B(B),
        .opcode(opcode),
        .result(result)
    );
    initial begin
        $dumpfile("../VCD/alu_8bit_updated.vcd");
        $dumpvars(0, tb_alu_8bit_updated);

        // Addition:
        A = 8'd20; B = 8'd10; opcode = 3'b000;
        #10;
        // Addition:  (8-bit result wraps to 0)
        A = 8'hFF; B = 8'h01; opcode = 3'b000;
        #10;
        // Subtraction:
        A = 8'd20; B = 8'd10; opcode = 3'b001;
        #10;
        // Subtraction: (8-bit underflow)
        A = 8'd5; B = 8'd10; opcode = 3'b001;
        #10;
        // AND
        A = 8'hF0; B = 8'h0F; opcode = 3'b010;
        #10;
        // OR
        A = 8'hF0; B = 8'h0F; opcode = 3'b011;
        #10;
        // XOR
        A = 8'hAA; B = 8'hFF; opcode = 3'b100;
        #10;
```

```
        // Left shift by 2
        A = 8'h0F; B = 8'd2; opcode = 3'b101;
        #10;

        // Left shift by 0
        A = 8'hA5; B = 8'd0; opcode = 3'b101;
        #10;

        // Right shift by 2
        A = 8'hF0; B = 8'd2; opcode = 3'b110;
        #10;

        // Right shift by 0
        A = 8'hA5; B = 8'd0; opcode = 3'b110;
        #10;

        // Default opcode
        A = 8'hFF; B = 8'hFF; opcode = 3'b111;
        #10;

        $finish;
    end

    initial begin
        $monitor("Time=%0t A=%h B=%h opcode=%b result=%h",
                 $time, A, B, opcode, result);
    end

endmodule
```

- **Commands:**

vcs -cm line+tgl+branch+cond -o Output/alu_8bit_updated.out ../RTL/Design/alu_8bit.v ../RTL/TB/tb_alu_8bit_updated.v

./alu_8bit_updated.out -cm line+tgl+branch+cond

urg -dir Output/alu_8bit_updated.out.vdb/ -report ../Coverage/

- **Updated Coverage:**

```
Time=40000 A=f0 B=0f opcode=010 result=00
Time=50000 A=f0 B=0f opcode=011 result=ff
Time=60000 A=aa B=ff opcode=100 result=55
Time=70000 A=0f B=02 opcode=101 result=3c
Time=80000 A=a5 B=00 opcode=101 result=a5
Time=90000 A=f0 B=02 opcode=110 result=3c
Time=100000 A=a5 B=00 opcode=110 result=a5
Time=110000 A=ff B=ff opcode=111 result=00
$finish called from file "../RTL/TB/tb_alu_8bit_updated.v", line 68.
$finish at simulation time              120000

------------------------------------------------------------------------
VCS Coverage Metrics: during simulation line, cond, branch, tgl was monitored
------------------------------------------------------------------------

 Coverage status: End of All Coverages ...

          V C S   S i m u l a t i o n   R e p o r t
Time: 120000 ps
CPU Time:      0.270 seconds;       Data structure size:   0.0Mb
Thu Oct 22 20:57:57 2026
[SCL] 10/22/2026 20:57:57 PID:317949 Client:smartlab6.calicut.nielit.in checkin
(null)
[oitp_b2-13@smartlab6 Output]$
```

SYNOPSYS

Dashboard

dashboard | hierarchy | modlist | groups | tests | asserts

Date: Thu Oct 22 21:00:43 2026
User: oitp_b2-13
Version: W-2024.09
Command line: urg -dir alu_8bit_updated.out.vdb/ -report ../Coverage/
Number of tests: 1

Total Coverage Summary

SCORE	LINE	COND	TOGGLE	BRANCH
95.23	98.65		98.15	88.89

Hierarchical coverage data for top-level instances

SCORE	LINE	COND	TOGGLE	BRANCH	NAME
95.23	98.65		98.15	88.89	tb_alu_8bit_updated

Total Module Definition Coverage Summary

SCORE	LINE	COND	TOGGLE	BRANCH
95.23	98.65		98.15	88.89

SYNOPSYS

Module Definition

dashboard | hierarchy | modlist | groups | tests | asserts

Module : alu_8bit

SCORE	LINE	COND	TOGGLE	BRANCH
92.35	90.00		98.15	88.89

Source File(s) :
/home/oitp_b2-13/Desktop/Shashaank_Malviya/Simulation/../RTL/Design/alu_8bit.v

Module self-instances :

NAME	SCORE	LINE	COND	TOGGLE	BRANCH
tb_alu_8bit_updated.uut	92.35	90.00		98.15	88.89

Module Instance : tb_alu_8bit_updated.uut

Instance :

SCORE	LINE	COND	TOGGLE	BRANCH
92.35	90.00		98.15	88.89

Instance's subtree :

SCORE	LINE	COND	TOGGLE	BRANCH
92.35	90.00		98.15	88.89

Line 〉 Toggle 〉 Branch 〉

Line Coverage for Module : alu_8bit

	Line No.	Total	Covered	Percent
TOTAL		10	9	90.00
ALWAYS	10	10	9	90.00

```
9                               always @(*) begin
10          1/1                     case (opcode)
11          1/1                         3'b000: result = A + B;       // Addition
12          1/1                         3'b001: result = A - B;       // Subtraction
13          1/1                         3'b010: result = A & B;       // Bitwise AND
14          1/1                         3'b011: result = A | B;       // Bitwise OR
15          1/1                         3'b100: result = A ^ B;       // Bitwise XOR
16          1/1                         3'b101: result = A << B[2:0]; // Logical left shi
17          1/1                         3'b110: result = A >> B[2:0]; // Logical right shi
18          1/1                         3'b111: result = 8'b00000000; // Unused opcode
19          0/1     ==>                 default: result = 8'b00000000;
```

Toggle Coverage for Module : alu_8bit

	Total	Covered	Percent